MEDIMATE MESSANGER
USING DECISION TREE ALGORITHM

TEAM MEMBERS

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ABSTRACT

The purpose of this paper is to show concisely how we can promote chatbots in the medical sector and cure infectious diseases. The chatbot harnesses the power of machine learning algorithms and a vast medical database to accurately recognize and name symptoms based on user input.The chatbot's primary functions include understanding the patient's description of their symptoms, identifying potential medical conditions associated with those symptoms, and recommending specialized doctors or medical specialists for further consultation. Users can interact with the chatbot through natural language conversations, making it user-friendly and accessible to individuals with varying levels of technological proficiency. Furthermore, the chatbot's functionalities extend beyond symptom recognition and doctor recommendations. It can also provide general information about medical conditions, first-aid advice, and preventive healthcare measures. Additionally, it respects patient privacy and complies with applicable data protection regulations, ensuring the secure handling of sensitive medical information. Result of this medical chatbot has the potential to alleviate the burden on healthcare systems by reducing unnecessary emergency room visits and facilitating early intervention for serious medical conditions.

AIM OF THE PROJECT

The aim of creating this Medimate Messenger (medical chatbot) using Decision Tree algorithm helps in enhancing the patients Healthcare Access and providing them Mental Health Support using Artificial Intelligence.

PROBLEM STATEMENT

The healthcare sector faces significant challenges in effectively diagnosing and treating infectious diseases while also managing the burden on healthcare systems. In light of these challenges, there is a need to explore innovative solutions that can promote chatbots in the medical sector to address the following problems:

- Inefficient Symptom Recognition
- Limited Access to Medical Specialists
- Lack of Public Health Education
- Privacy Concerns

This project is focused to improve healthcare access for patients and offer mental health support through the application of artificial intelligence (AI).

LITERATURE SURVEY

S. Chakraborty and L. Dey, ‘‘The implementation of AI and AI-empowered imaging system to fight against COVID-19—A review,’’ in Smart Health-care System Design: Security and Privacy Aspects. 2022, pp. 301–311.
Covid-19 is an ailment because of the SARS-CoV-2 virus. World Health Organization (WHO)’ has been declared a pandemic on March 11, 2020. Around 15 million human beings have been affected worldwide by more than one mil-lion deaths by covid-19.The purpose of this paper is to show concisely how we can promote chatbots in the medical sector and cure infectious diseases. We can create awareness through the users and the users can get proper medical solutions to prevent disease. We created a preliminary training model and a study report to improve human interaction in databases in 2021.
When someone gets severe Covid-19,the oxygen levels in the body get low. Patients suffered from enormous fevers, coughs, and lost sense of taste and smell, all of which became problematic when the SARS-CoV-2 virus invaded them.

J. Cahn, ‘‘CHATBOT: Architecture, design, & development,’’ Dept. Com-put. Inf. Sci., Univ. Pennsylvania School Eng. Appl. Sci., Philadelphia, PA,USA, Tech. Rep. EAS499, 2017.
The reference you provided is for a technical report titled "CHATBOT: Architecture, design, & development" authored by J. Cahn. This report was produced as a part of the Department of Computer and Information Science at the University of Pennsylvania School of Engineering and Applied Science in Philadelphia, PA, USA.

While the details of the report are not provided in your text, it's likely that this technical report covers the architecture, design, and development aspects of chatbots. It may provide insights into the technical and practical aspects of creating chatbot systems. If you have access to this report, you can refer to it for in-depth information on chatbot development.

E. Tebenkov and I. Prokhorov, ‘‘Machine learning algorithms for teaching AI chat bots,’’ Proc. Comput. Sci., vol. 190, pp. 735–744, Jan. 2021.

The reference you provided is for a paper titled "Machine learning algorithms for teaching AI chat bots" authored by E. Tebenkov and I. Prokhorov. This paper was published in the Proceedings of Computer Science, volume 190, and it appears to have been published in January 2021.

While the details of the paper are not provided in your text, it's likely that this paper discusses the use of machine learning algorithms in the training and development of AI chatbots. It may cover various aspects of machine learning techniques and methods for improving the capabilities of chatbot systems. If you have access to the paper, you can refer to it for a more in-depth understanding of the topic.

R. Dharwadkar and N. A. Deshpande, ‘‘A medical chatbot,’’ Int. J. Comput. Trends Technol., vol. 60, no. 1, pp. 41–45, 2018.

The reference you provided is for a paper titled "A medical chatbot" authored by R. Dharwadkar and N. A. Deshpande. This paper was published in the International Journal of Computer Trends and Technology, in volume 60, issue 1, and it appears to have been published in 2018.

Although the specific content of the paper is not provided in your text, the title suggests that it likely discusses the development and application of a medical chatbot. It may cover the design, implementation, and potential uses of a chatbot in the field of healthcare. If you have access to the paper, you can read it to gain a more detailed understanding of the topic.

EXISTING SYSTEM

1**.Objective:** This project aims to demonstrate the promotion of chatbots in the medical sector to combat infectious diseases.
2.User Awareness: The authors suggest that creating user awareness can lead to proper medical solutions and disease prevention through chatbots.
3.Preliminary Training Model: A training model and study report were created in 2021 to improve human interaction in databases, particularly in the context of chatbots and natural language processing.
4.Human Behavior Analysis: The study employs natural language processing to describe human behaviors and characteristics of the chatbot.
5.AI Chatbot Interaction Model: The paper proposes an AI Chatbot interaction and prediction model using a deep feedforward multilayer perceptron.
6.Functionality and Applications: The research explores the functionalities and potential applications of medical chatbots, particularly in addressing health crises like pandemics such as COVID-19.

PROPOSED SYSTEM

1.24/7 Availability : Healthcare chatbots offer 24/7 assistance, addressing various inquiries from general to complex medical questions for patients and potential clients whenever they visit the website.

2.Appointment Booking: Chatbots facilitate the booking of appointments with healthcare professionals, streamlining the process for patients and reducing the administrative burden on staff.

3.Information Delivery: They can automatically provide relevant information to potential patients, ensuring they have access to the right resources and details about the hospital's services. It can also provide educational resources, such as articles or videos, to help patients better understand their medical conditions and treatment options.

4.Cost-Efficiency: Chatbots can handle a high volume of inquiries simultaneously, reducing the need for additional human resources and making healthcare services more cost-effective.

5.Privacy and Security: Healthcare chatbots are designed with strict adherence to privacy and security regulations, ensuring the protection of sensitive patient data.

6.Patient Engagement and Reduced Wait Times: They can engage patients by sending reminders for appointments, follow-up care, and medication schedules, enhancing patient adherence to treatment plans. Patients do not have to wait on hold or in queues, as chatbots can quickly address their queries and concerns.

7.Enhanced Efficiency: By automating routine tasks and inquiries, healthcare chatbots allow human staff to focus on more complex and critical aspects of patient care.

8.User Feedback and Improvement: They can collect user feedback and data, which can be used to continuously improve the chatbot's responses and user experience.

SOFTWARE REQUIREMENTS:

1. A database system for storing user interactions and data, such as CSV File
2. Python version (3.7 or any other version)
3. Chrome or any other Web Browser
4. Sublime Text Code Editor
5. Libraries Required:
    - Numpy
    - Scikit Learn
    - Pandas
    - Matplotlib
    - Tkinter

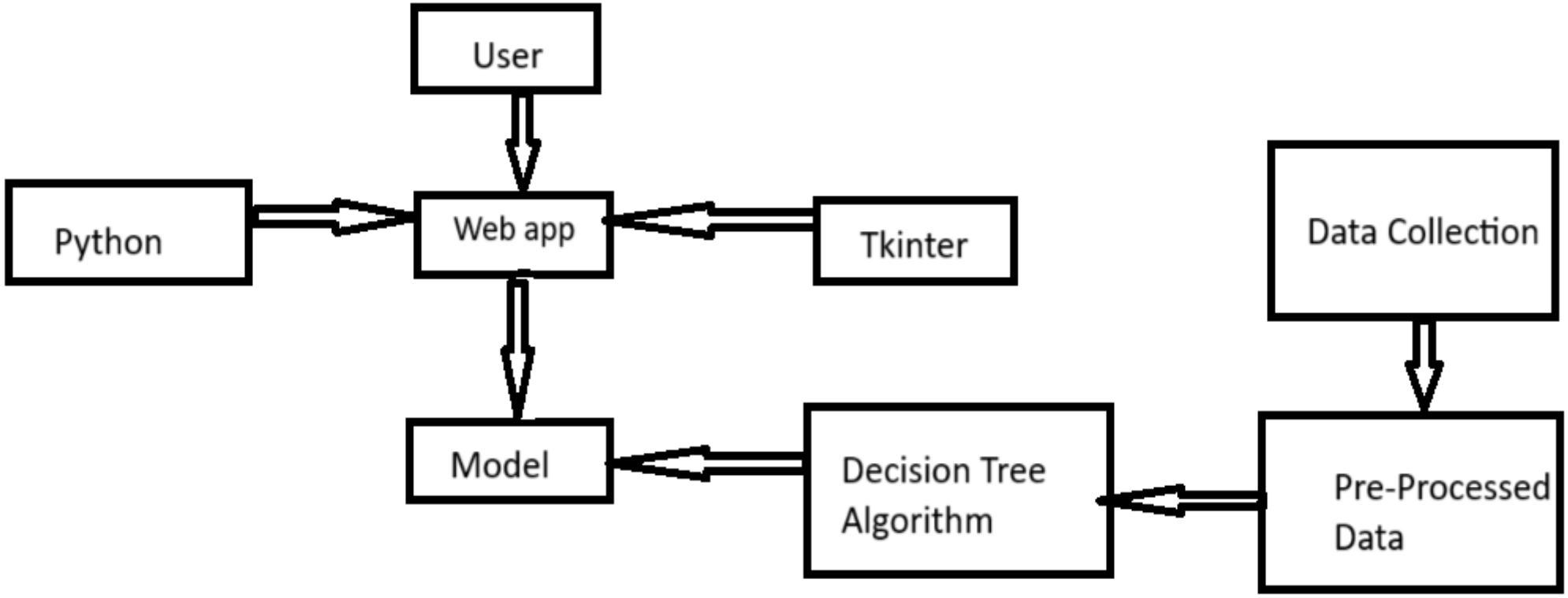
PYTHON FRAME WORK:

1.Tkinter

HARDWARE REQUIREMENTS:

1.Windows 10
2.Processor –i3
3.RAM-4gb or above
4.ROM-512SSD

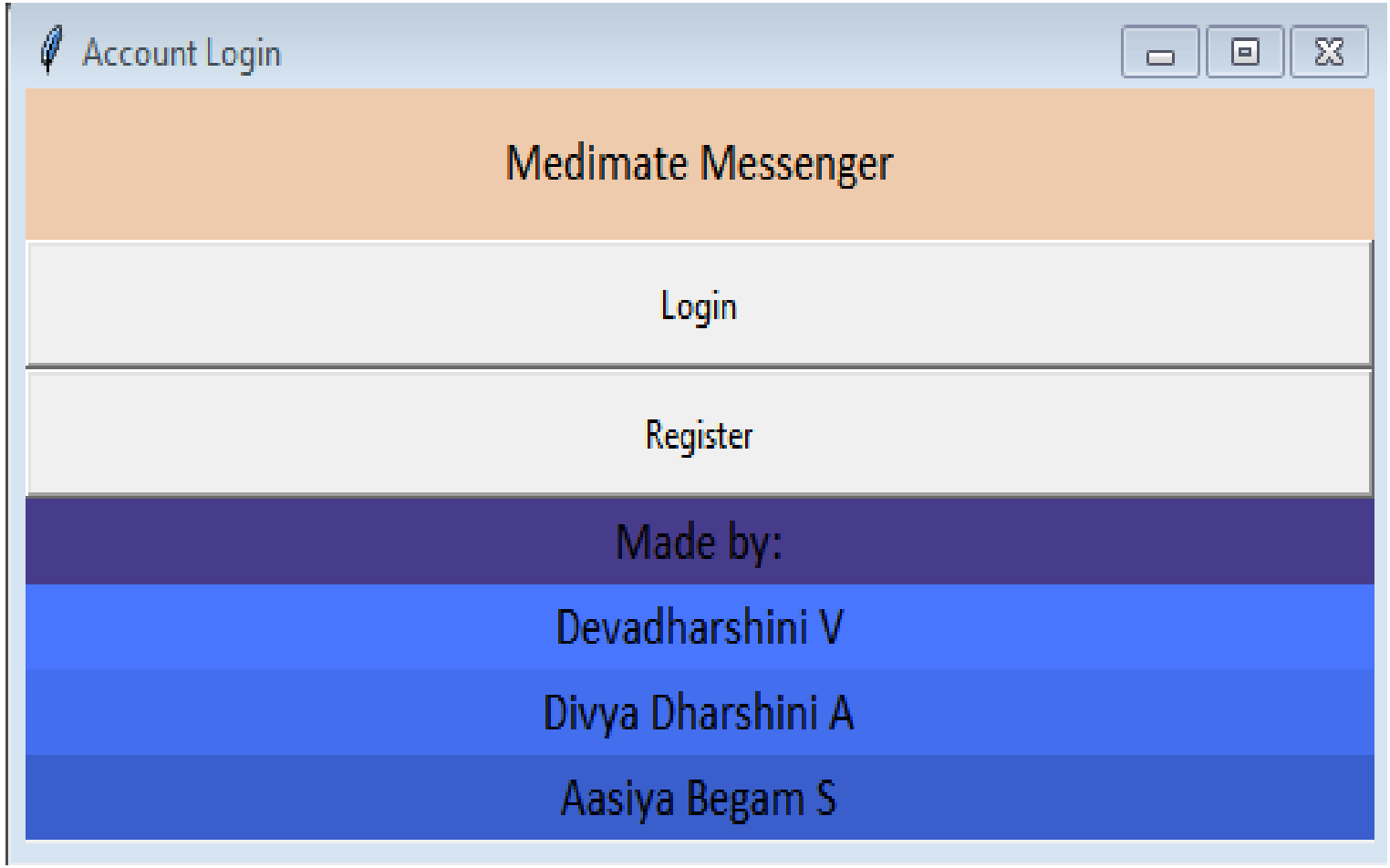
DATA FLOW DIAGRAM



MODULES

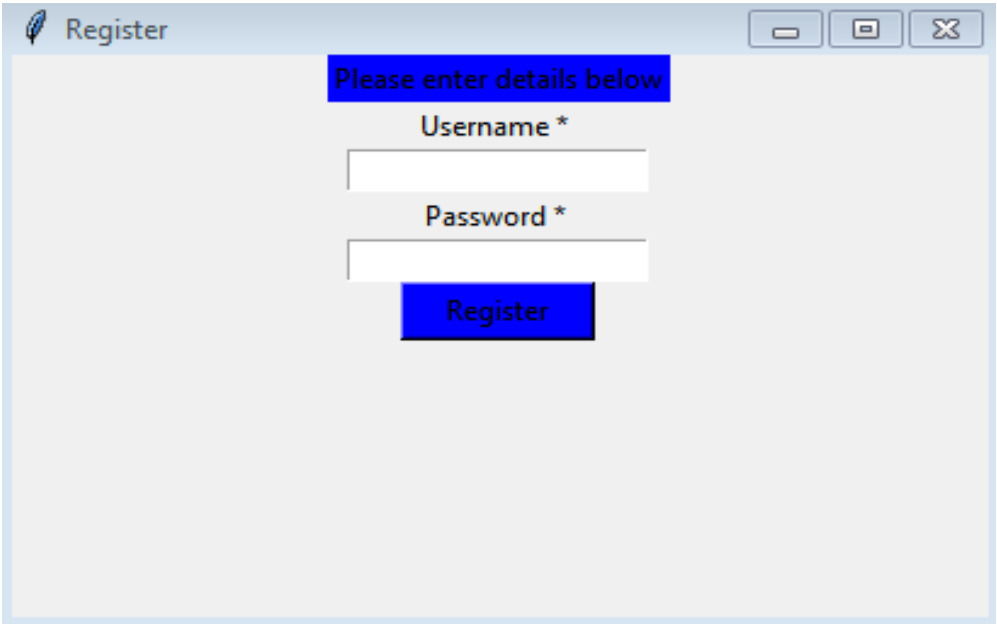
- Installation module
- Environmental setup module
- Data collection and pre processing module
- Implementing the Decision Tree Algorithm
- Register and login module
- Training the model
- Testing the model
- Model evaluation
- Web app integration using Tkinter

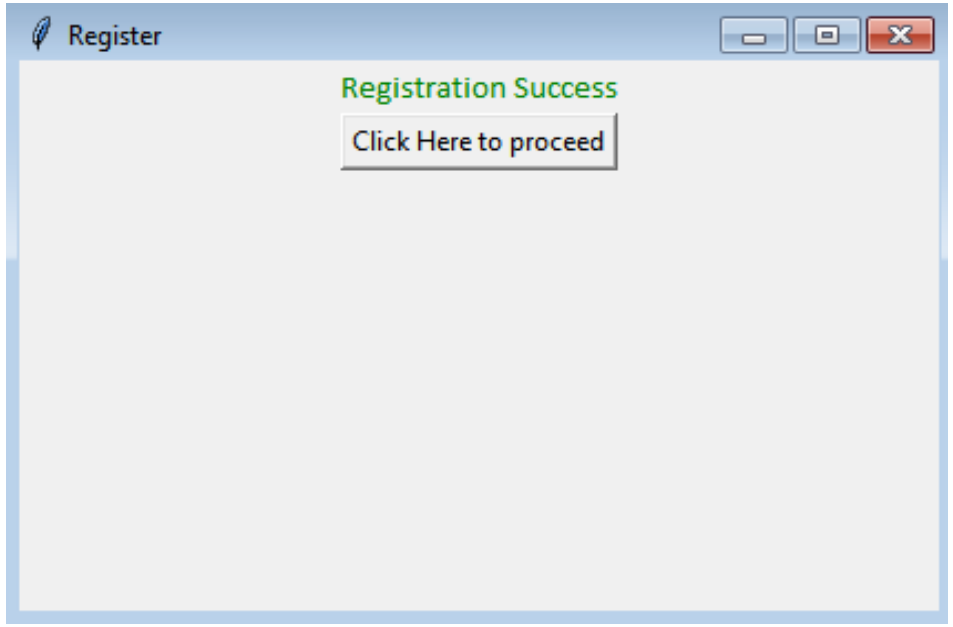
SCREENSHOT



Main page

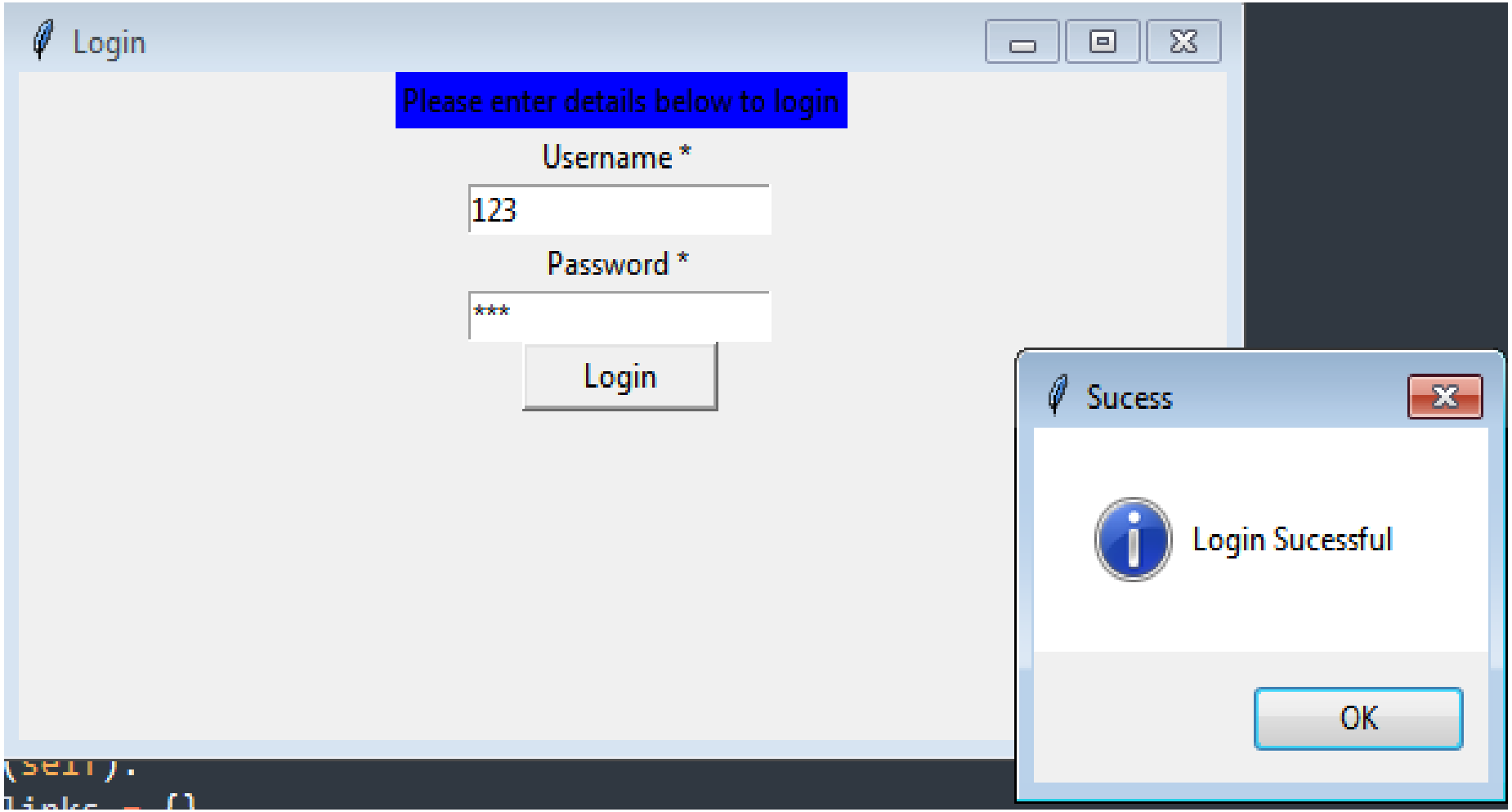
SCREENSHOT





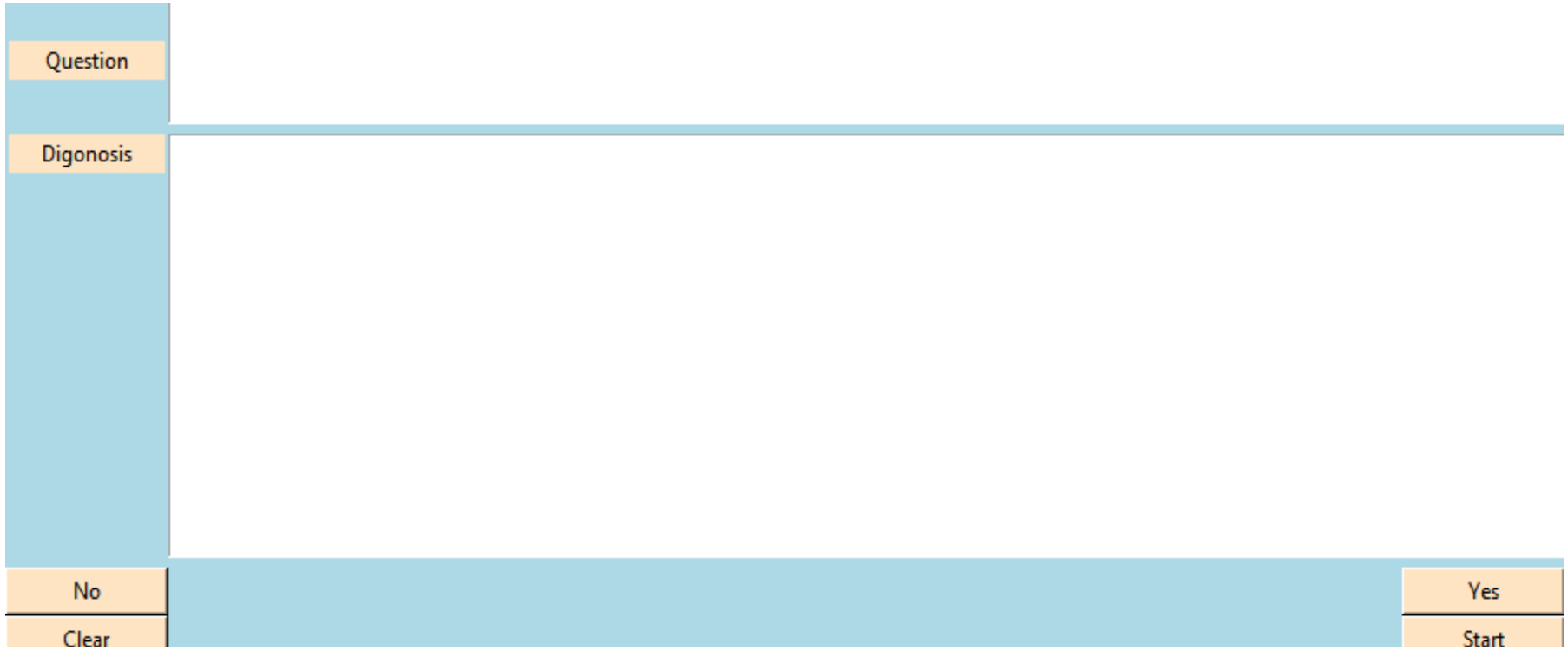
Register Page

SCREENSHOT



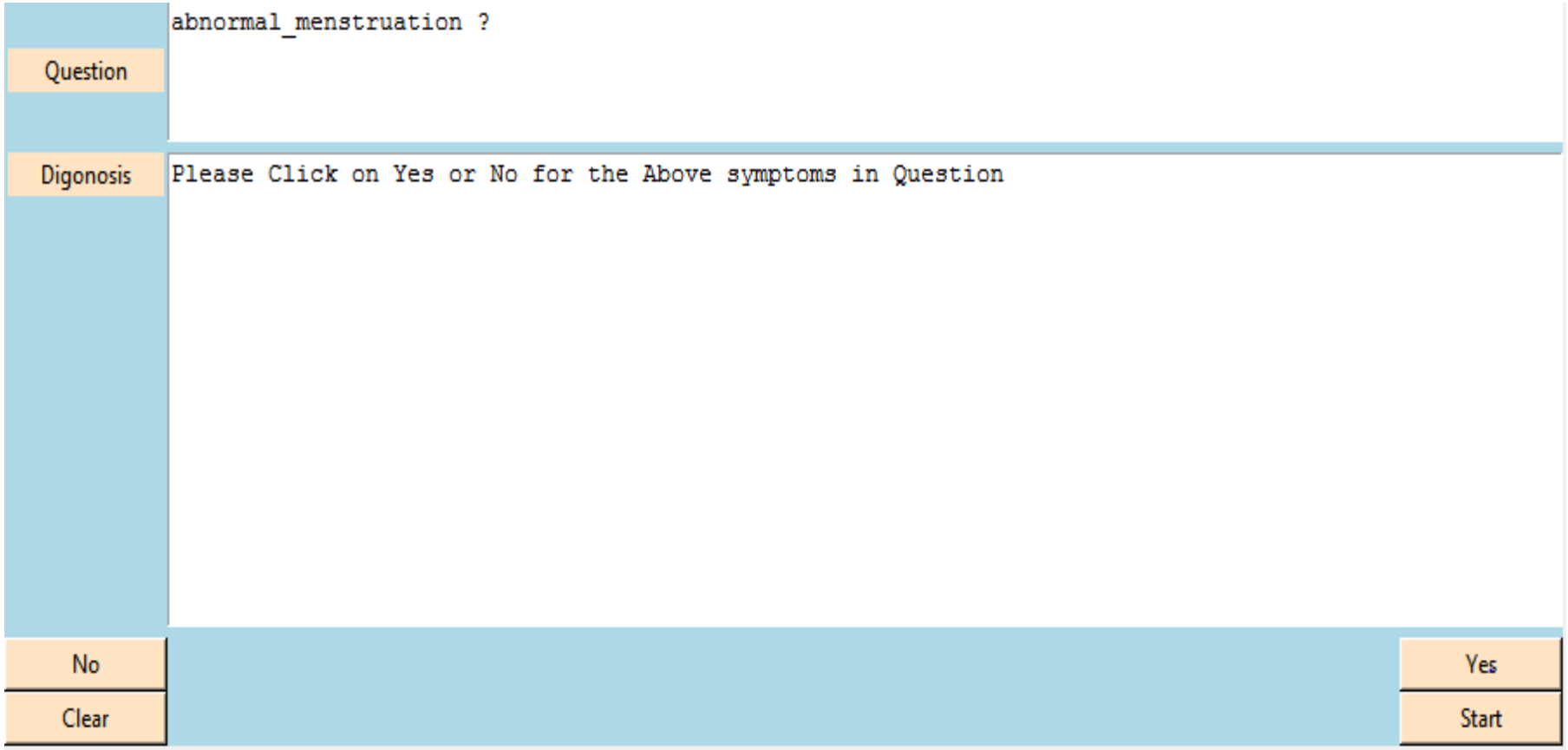
Login Page

SCREENSHOT



1. Click the start button

SCREENSHOT



2. Click Yes or No button, based on your symptoms.

OUTPUT

Question

skin_rash ?

Digonosis

You may have :['Dengue']
symptoms present: ['skin_rash', 'dischromic _patches']
symptoms given: ['skin_rash', 'chills', 'joint_pain', 'vomiting', 'fatigue', 'high_fever', 'headache', 'nausea', 'loss_of_appetite', 'pain_behind_the_eyes', 'back_pain', 'malaise', 'muscle_pain', 'red_spots_over_body']
confidence level is: 0.14285714285714285
The model suggests:
Consult ['Dr. Ajay Jain']
Visit https://www.practo.com/delhi/doctor/dr-ajay-jain-ear-nose-throat-ent-specialist-1?specialization=Ear-Nose-Throat%20(ENT)%20Specialist&practice_id=664069

No

Clear

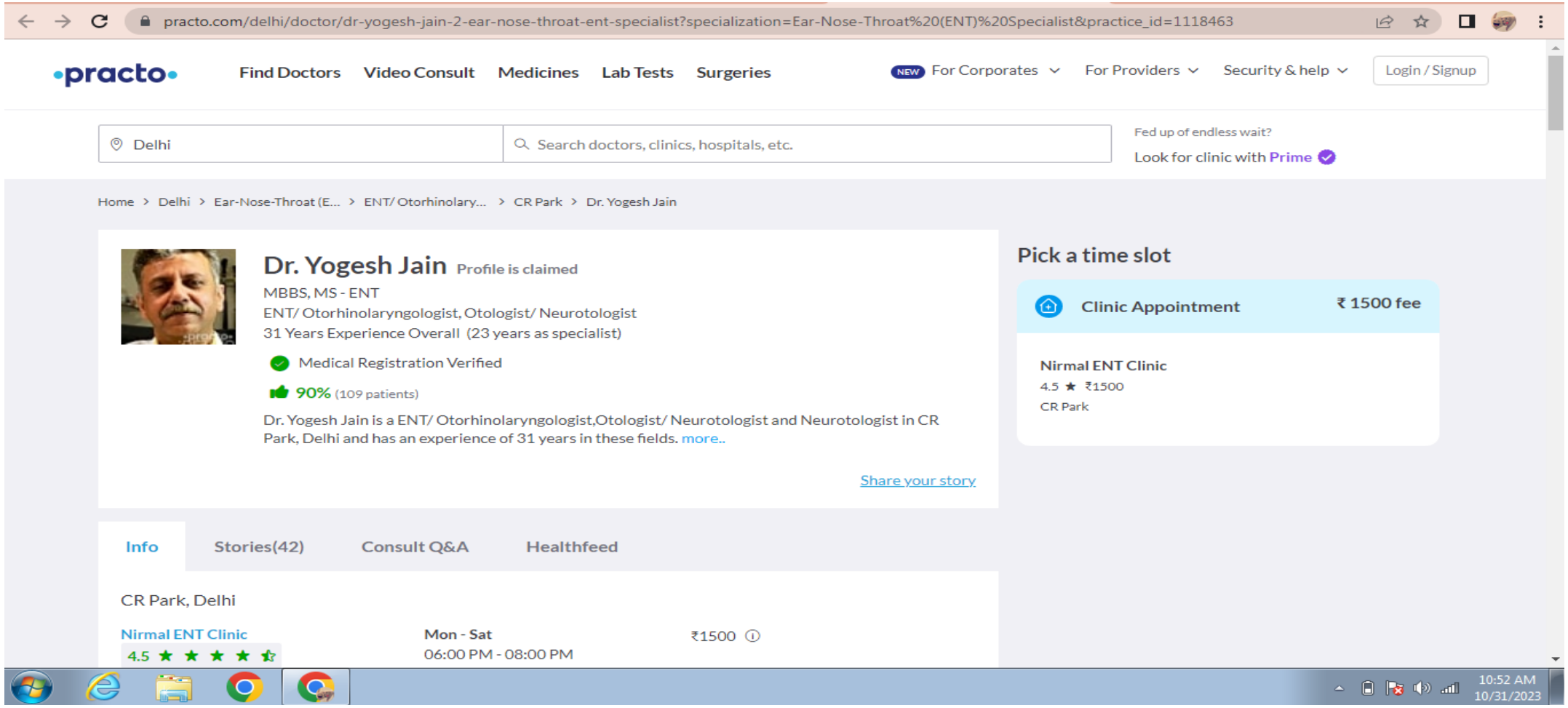
Yes

Start

Doctor Suggestion Page

3.Click the visit link to book your appointment.

OUTPUT



Doctor's Appointment Booking Page

CONCLUSION

The development of a Medical Chatbot that combines symptom recognition, doctor recommendations, and AI-driven treatment guidance signifies a promising evolution in the healthcare industry. This technology not only facilitates better healthcare accessibility but also empowers individuals to make informed decisions about their health. As healthcare continues to evolve, such solutions serve as a testament to the potential of AI in improving patient outcomes and making healthcare more patient-centric and efficient. Result of this medical chatbot has the potential to alleviate the burden on healthcare systems by reducing unnecessary emergency room visits and facilitating early intervention for serious medical conditions.

FUTURE ENHANCEMENT

The model is only accessable via webapp, in future we will bw extending this app for mobile application and also introducing some other features and model alteration by the user.

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Thank You